你说：

Please check this text for grammatical and punctuation errors, reply in English:

```
% ================================================
% 1. Introduction (5 points)
% ================================================
\section{Introduction}

\textit{Can we anticipate military escalation from major power?} As one of the central questions in conflict research, the answer concerns not only a broad scholarship in international relations and comparative politics, but also policymaking and diplomatic decisions.

As with other forecasting tasks, machine learning models have been widely deployed on this question. Starting from the ground-breaking work of \citet{beck2000improving}, which used neural networks to predict militarized interstate disputes, time-series models \citep{brandt2011real}, random forests \citep{muchlinski2016comparing}, and gradient boosting \citep{hegre2021views2020} were successively introduced.

A second line of work focuses on the training data itself, moving from structural variables such as conflict history, population, development, and geography \citep{hegre2021can} to unstructured data. A representative direction is the rise of text-as-data. \citet{chadefaux2014early} first introduced news text into war early-warning, while \citet{mueller2018reading} further employed topic models to extract information from newspaper articles, systematically incorporating media information into conflict forecasting. More recently, \citet{wong2026} extended this line of research by exploring the predictive value of official rhetoric.

Following this line of research, this project seeks to advance military escalation forecasting by extending the analysis of unstructured data beyond text. As a form of tacit knowledge \citep{polanyi1966tacit}, images are often more effective in shaping emotions \citep{brader2005striking, slovic2017iconic}. \citet{powell2015clearer} shows that when text and images appear together, textual frames shape opinions, while visual frames drive behavioral intentions. Images may therefore play a distinctive role in wartime mass mobilization and legitimacy-building \citep{mattingly2022soft}, offering a potential avenue for improving military escalation forecasts.

% On the one hand, media information may be biased. As \citet{cederman2017predicting} point out, internet access in many high-risk regions is censored or restricted to a small segment of the population. Consequently, the predictive value of textual data may depend on media freedom. On the other hand, other forms of unstructured data, particularly images, convey information beyond text.



Building on these, we compile an original dataset of 5,619 Chinese propaganda posters spanning the 1950s to the 2010s. By extracting
```

visual and textual features, we explore their potential to improve forecasts of military escalation.

…

显示更多

ChatGPT 说：

Your introduction is generally clear and grammatically sound. I found a few grammatical errors, awkward constructions, and minor stylistic inconsistencies. Below is a **minimally revised version** that preserves your original meaning, academic tone, and LaTeX citations.

1. Corrected version

```latex
\section{Introduction}

\textit{Can we anticipate military escalation by major powers?} This question is central to conflict resear

As with other forecasting tasks, machine learning models have been widely applied to this question. Beginni

A second line of work focuses on the training data itself, moving from structural variables such as conflic

Following this line of research, this project seeks to advance military escalation forecasting by extending

Building on this literature, we compile an original dataset of 5,619 Chinese propaganda posters spanning th
```

2. Specific corrections

Original	Suggested correction	Reason
from major power	by major powers	Missing article/plural; *by* more naturally identifies the actor conducting escalation.
As one of the central questions ..., the answer concerns ...	This question is central to ..., with implications ...	Original has an illogical grammatical relationship: the *answer* is not a question.
deployed on this question	applied to this question	More idiomatic academic English.
Starting from ... models ... were successively introduced	Beginning with ... subsequent studies introduced ...	Corrects the awkward introductory modifier and sentence structure.
ground-breaking	groundbreaking	Both are acceptable; the latter is more common in American English.
A representative direction is the rise of text-as-data	One prominent direction is the use of text as data	More natural academic phrasing.
text-as-data. \citet	text as data. \citet	Removes double spacing.
war early-warning	war early warning	Avoids an unnecessary hyphen when the phrase functions as a noun.
more effective in shaping emotions	more effective at shaping emotions	More idiomatic construction.
Building on these	Building on this literature	Clarifies the antecedent of *these*.

3. Additional academic writing observations

These are not strictly grammatical errors, but three expressions deserve consideration.

1. "As a form of tacit knowledge"

This is grammatically correct, but the conceptual relationship between images and tacit knowledge is not entirely self-evident. Polanyi's concept of tacit knowledge does not necessarily imply that images themselves constitute tacit knowledge. You may want to clarify the theoretical connection.

2. "Images are often more effective at shaping emotions"

The comparative *more effective* implicitly raises the question: more effective than what?

If you are explicitly comparing images with text, you could write:

> Images are often more effective than text at eliciting emotional responses.

This is more precise, provided that your cited literature supports the comparison.

3. "Military escalation" versus "militarized interstate disputes"

Your opening emphasizes *military escalation*, while your empirical discussion concerns forecasting militarized interstate disputes. These are related but not equivalent concepts. If your outcome is the annual number of active disputes, consider whether *military conflict forecasting* or *forecasting militarized interstate disputes* would describe the empirical task more precisely.

Overall: The most important grammatical problems are in the opening sentence, the construction beginning with *Starting from*, and *Building on these*. The remaining corrections are primarily idiomatic or stylistic. Your paragraph organization and transitions are already coherent, and no substantial restructuring is necessary.

你说：

Please check this text for grammatical and punctuation errors, reply in English: %
```
================================================
% 2. Data (10 points)
% ================================================
\section{Data}
Two main datasets were applied: Posters collected from
\href{https://chineseposters.net}{\textcolor[rgb]{0,0.3,0.6}
{ChinesePosters.net}} through a WebCrawler, and the Militarized
Interstate Disputes (MIDs) dataset, from the Correlates of War Project
\citep{palmer2022mid5}.


\subsection{Propaganda Posters}

We crawled 5,619 unique poster records from the Chinese Posters
Foundation. The collected records contain poster images with Chinese
and English titles, publication dates, publishers, designers, physical
dimensions, website descriptions, themes, and tags. After parsing
publication dates, we retain 4,752 posters with unambiguous
publication years between 1949 and 2013.

For data preprocessing, we first deal with color features by applying
EXIF orientation correction and converted images with embedded color
profiles to sRGB. We calculate mean RGB values, mean HSV
saturation, and CIELAB lightness statistics for each poster using scikit-
image \citep{vanderwalt2014skimage}. Meanwhile, we identify red
pixels through threshold-based segmentation in HSV color space
\citep{opencv_hsv_threshold}. Specifically, we define red pixels as
having a hue within $15^\circ$ of red, saturation $\geq 0.4$, and value
$\geq 0.15$. This procedure is illustrated in following example:

After extracting the color features, we standardize the recorded physical
dimensions by correcting clear formatting inconsistencies, and after that
We derive the shorter side, longer side, printed area, and aspect ratio.
This process retains some unsupported or ambiguous formats which
we save as missing values. We also extract Chinese-title availability
and Chinese character counts. For designers, we count identified
named entities and create an indicator for multiple entities.

Website description lengths, theme and tag counts, and selected
subject indicators form a separate feature group. While these
retrospective annotations provide additional information, they may also
```

introduce data leakage in temporal prediction. We therefore conservatively retain only two military-related tags.

We retain observations with missing feature values rather than dropping them. To better align with the target variable, we aggregate features into annual means.\footnote{For binary indicators, annual means represent proportions among non-missing records.}

\subsection{Military Escalation}

Following the established literature on military escalation prediction, we use the Dyadic Militarized Interstate Disputes dataset, version 4.03. \citep{beck2000improving, palmer2022mid5}. We focus on records involving China (country code 710) from 1950 to 2014.

Our target variable is the annual number of distinct disputes in which China is involved. As a single dispute can involve multiple countries, we count each dispute only once per year, and we count ongoing disputes in every year they remain active. The outcome thus captures China's annual participation in militarized disputes. Matching poster features from year t to dispute counts in year $t+1$ yields 65 annual observations.

\subsection{Dataset Overview}

The final dataset contains 65 rows and 125 columns, including identifiers, aggregated features, data-availability indicators, the target variable, and the split label. The unit of observation is a calendar year. Poster features from year \(t\) are matched to a single count target: the number of distinct militarized interstate disputes involving China that are active in year \(t+1\). Detailed feature descriptions are provided in the appendix and in the data documentation in our \href{https://github.com/Kaidi-Zhu-DS/Posters.git}{\textcolor[rgb]{0,0.3,0.6}{Git Repository}}.
…

显示更多

ChatGPT 说：

Your Data section is generally clear and well organized. I found several grammatical errors, punctuation issues, and awkward expressions, particularly in the preprocessing paragraph. Below is a **minimally revised version** that preserves your original meaning, technical details, and LaTeX formatting.

1. Corrected version

LaTeX

```
% ================================================
% 2. Data (10 points)
% ================================================
\section{Data}

Two main datasets were used: posters collected from \href{https://chineseposters.net}{\textcolor[rgb]{0,0.3


\subsection{Propaganda Posters}

We crawled 5,619 unique poster records from the Chinese Posters Foundation. The collected records contain p
```

```
For data preprocessing, we first process color features by applying EXIF orientation correction and convert

After extracting the color features, we standardize the recorded physical dimensions by correcting clear fo

Website description lengths, theme and tag counts, and selected subject indicators form a separate feature

We retain observations with missing feature values rather than dropping them. To better align with the tar
```

2. Specific corrections

Original	Correction	Reason
Two main datasets were applied	Two main datasets were used	*Used* is the more natural verb for datasets.
Posters collected	posters collected	No capitalization needed after a colon here.
through a WebCrawler	using a web crawler	Correct capitalization and conventional terminology.
dataset, from the Correlates	dataset from the Correlates	Unnecessary comma.
contain poster images with Chinese and English titles	contain poster images, along with Chinese and English titles	Clarifies that titles are metadata rather than elements within the images.
by applying ... and converted images	by applying ... and converting images	Corrects inconsistent verb forms.
we first deal with color features	we first process color features	More natural in a technical methodology section.
illustrated in following example	illustrated in the following example	Missing definite article.
and after that We derive	We then derive	Fixes capitalization and an awkward sentence construction.
which we save as missing values	We record ... as missing values	Clarifies the treatment of unsupported formats.
version 4.03. `\citep{...}`.	version 4.03 `\citep{...}`.	Removes the extra period and correctly positions the citation.
every year they remain active	every year in which they remain active	Slightly more formal phrasing.
Git Repository	GitHub repository	Correct term and capitalization for a repository hosted on GitHub.

3. Additional observations

Beyond grammatical corrections, I would consider three methodological wording issues.

1. "This procedure is illustrated in the following example:"

The colon is correct if a figure or example immediately follows. If you are introducing a figure in LaTeX, however, the more conventional construction is:

> This procedure is illustrated in Figure \ref.

This avoids leaving a sentence ending in a colon without a visible continuation.

2. "We therefore conservatively retain only two military-related tags."

The sentence is grammatically correct, but its relationship to data leakage could be clearer. Because your concern is retrospective annotations, retaining two military-related tags does not necessarily eliminate leakage.

If the two tags were selected based on substantive relevance, rather than because they were known at the prediction date, I would retain your wording but avoid suggesting that this selection fully addresses temporal leakage.

3. "Military escalation" versus "annual participation in militarized disputes"

As in your Introduction, there is a conceptual distinction between escalation and the number of active disputes. Your target variable captures the **annual count of active militarized interstate disputes involving China**, not necessarily the escalation of military conflict.

The current description of your target variable is accurate and precise. I would simply ensure that your broader terminology remains consistent throughout the paper.

Overall assessment: The clearest grammatical problems were the inconsistent verb forms (*applying ... and converted*), the missing article (*in following example*), the capitalization error (*after that We*), and the misplaced citation punctuation. The remaining changes primarily improve academic English, readability, and precision.

你说：

Please check this text for grammatical and punctuation errors, reply in English: %
==========
% 3. Exploratory Data Analysis (10 points)
% ==========
\section{Exploratory Data Analysis}

Given the temporal structure of the data, we chronologically divide the final 65 observations into 52 training and validation observations (poster years 1949–2000) and 13 held-out test observations (2001–2013). All EDA in this section uses only the training and validation data.

\subsection{From 25 Candidates to 12 Features}

To ensure broad representation across different types of information, we begin by identifying 25 theoretically informed candidate features from the original dataset, covering three main categories:

\begin{itemize}
  \item \textbf{Visual:} Features directly extracted from poster images, including red, green, and blue color measures, lightness, saturation, and other visual characteristics.
  \item \textbf{Metadata:} Information associated with the posters but not directly embedded in the images, including titles, designer characteristics, and publisher attributes.
  \item \textbf{Archive:} Subjective annotations added by the collectors or archivists. To minimize potential biases introduced by these retrospective classifications, we conservatively selected only two military-related indicators.
\end{itemize}

Given that 25 features may lead to overfitting with limited training observations, we examined feature variability, pairwise Pearson correlations, conceptual overlap, and relevance to the research question to reduce the feature set. We retained one representative from several highly overlapping groups. The full correlation matrix and the 13 exclusion decisions appear in Appendix~\ref{fig:correlations}.

We also excluded nearly constant or particularly sparse name-based indicators. These exclusions were exploratory choices, and we will carefully consider how to incorporate these features and better handle omitted variables in the subsequent model training stage.

Table~\ref{tab:selected_features} lists the final provisional selection.

\begin{table}[H]
\centering
\caption{Selected features for the proposed regression model.}
\label{tab:selected_features}

```
\footnotesize
\setlength{\tabcolsep}{4pt}
\renewcommand{\arraystretch}{1.2}

\begin{tabularx}{\textwidth}{
   @{}p{1.4cm}p{3.5cm}YY@{}
}
\toprule
\textbf{Category} &
\textbf{Feature} &
\textbf{Description} &
\textbf{Selection Rationale} \\
\midrule

Visual &
\nolinkurl{red_ratio_mean} &
Mean proportion of red pixels &
Captures red-color prevalence \\

Visual &
\nolinkurl{saturation_mean_mean} &
Mean color saturation &
Measures color vividness \\

Visual &
\nolinkurl{lightness_mean_mean} &
Mean image lightness &
Represents overall brightness \\

Visual &
\nolinkurl{lightness_std_mean} &
Mean within-image lightness standard deviation &
Captures brightness variation \\

Visual &
\nolinkurl{size_area_cm2_mean} &
Mean printed area ($\mathrm{cm}^2$) &
Represents physical poster size \\

Visual &
\nolinkurl{size_aspect_ratio_mean} &
Mean poster aspect ratio &
Captures poster shape \\

\midrule

Metadata &
\nolinkurl{title_han_count_mean} &
Mean Chinese character count in titles &
Measures title length \\

Metadata &
\nolinkurl{designer_has_identified_mean} &
Proportion with identified designers &
Captures designer attribution \\

Metadata &
\nolinkurl{designer_org_name_marker_mean} &
Proportion with organizational designer-name markers &
```

```
Captures organizational attribution \\

Metadata &
\nolinkurl{publisher_military_name_marker_mean} &
Proportion with military publisher-name markers &
Captures military-related publishing \\

\midrule

Archive &
\nolinkurl{archive_theme_military_mean} &
Proportion with military-theme labels &
Captures military-themed content \\

Archive &
\nolinkurl{archive_tag_military_personnel_mean} &
Proportion with military-personnel tags &
Captures military personnel references \\

\bottomrule
\end{tabularx}

\end{table}

\subsection{Visualizations and Explanatory Findings}

As shown in Figure~\ref{fig:dispute}, the annual number of active MIDs varies considerably across the study period. Notably, the average annual dispute count declined from 5.0 during 1950–1969 to 2.3 during 1970–1999. This historical variation suggests that the underlying relationships may change over time, motivating chronological rather than random splits.

\begin{figure}[H]
  \centering
  \caption{Decline in Military Disputes Involving China}
  \label{fig:dispute}
  \includegraphics[width=0.9\linewidth]{01_dispute_counts.png}
  \caption*{\small \textit{Note:} The left panel displays annual dispute counts, while the right panel shows their distribution across the training set. Average counts were higher during 1950–1969 than during 1970–1999.}
\end{figure}
\vspace{-20pt}

\begin{figure}[H]
  \centering
  \caption{Declining Poster Production}
  \label{fig:availability}
  \includegraphics[width=0.9\linewidth]{06_archive_coverage.png}
  \caption*{\small \textit{Note:} Yearly number of eligible posters and percentage with a valid military-theme annotation.}
\end{figure}
\vspace{-20pt}

Figure~\ref{fig:availability} suggests that archive availability is uneven. Overall, however, a potential positive correlation can be observed: as China's involvement in militarized disputes declined, the production of military-related posters also appears to have decreased.
```

Military markers show broadly similar patterns. Figure~\ref{fig:military} displays two indicators: military publishers and organizational designers tag, and two archival annotations: military themes and personnel tag. Their peaks coincide with or precede heightened disputes, indicating propaganda may signal military escalation via mass mobilizing.

% \begin{figure}[H]
% \centering
% \includegraphics[width=1\linewidth]
{04_military_metadata_and_archive_trends.png}
% \caption{Three aligned time-series panels. Vertical scales reflect shares among records with known values and may differ
% between panels.}
% \label{fig:military}
% \end{figure}

\begin{figure}[H]
 \centering
 \caption{Coinciding Peaks in Military-Related Poster Indicators and Military Disputes}
 \label{fig:military}
 \includegraphics[width=1\linewidth]
{04_military_metadata_and_archive_trends.png}
 \caption*{\small \textit{Note:} Three aligned time-series panels. Vertical scales reflect shares among records with known values and may differ between panels.}
\end{figure}
\vspace{-20pt}

Visual properties, however, present more mixed evidence. Figure~\ref{fig:visual} compares red-pixel share, saturation, lightness, and printed area. Earlier posters generally exhibit higher saturation, while later posters tend to have larger printed areas. Their relationship with military escalation, however, leaves room for more nuanced exploration.

\begin{figure}[H]
 \centering
 \caption{Distinct Visual Styles Across Historical Periods but Mixed Evidence in Disputes}
 \includegraphics[width=1\linewidth]{02_visual_trends.png}
 \caption*{\small \textit{Note:}{Four annual visual summaries. Moving averages are descriptive. Faint curves indicate observed yearly means; thick curves show centered five-year descriptive averages. Units are shown on each axis.}}
 \label{fig:visual}
\end{figure}
…

显示更多

ChatGPT 说：

Your Exploratory Data Analysis section is generally clear and well structured. I found several grammatical errors, awkward constructions, and minor punctuation issues, especially in the discussion of military-related indicators and the final paragraph.

Below is a **minimally revised version** that preserves your original argument, technical terminology, and LaTeX structure.

1. Corrected version

```latex
% ================================================
% 3. Exploratory Data Analysis (10 points)
% ================================================
\section{Exploratory Data Analysis}

Given the temporal structure of the data, we chronologically divide the final 65 observations into 52 train

\subsection{From 25 Candidates to 12 Features}

To ensure broad representation across different types of information, we begin by identifying 25 theoretica

\begin{itemize}
    \item \textbf{Visual:} Features directly extracted from poster images, including red, green, and blue c
    \item \textbf{Metadata:} Information associated with the posters but not directly embedded in the image
    \item \textbf{Archive:} Subjective annotations added by collectors or archivists. To minimize potential
\end{itemize}

Given that 25 features may lead to overfitting with a limited number of training observations, we examine f

We also exclude nearly constant or particularly sparse name-based indicators. These exclusions are exploi
```

2. Specific corrections

Original	Suggested correction	Reason
added by the collectors or archivists	added by collectors or archivists	More natural when referring to collectors and archivists in general.
we conservatively selected	we conservatively select	Maintains consistent present tense in the methodology.
with limited training observations	with a limited number of training observations	More idiomatic expression.
we examined feature variability	we examine feature variability	Maintains tense consistency.
We retained one representative from several highly overlapping groups	We retain one representative feature from each of several highly overlapping groups	Clarifies what was retained and from which groups.
the 13 exclusion decisions appear	the decisions to exclude 13 features are presented	More natural academic expression.
We also excluded	We also exclude	Tense consistency.
These exclusions were exploratory choices	These exclusions are exploratory choices	Tense consistency.
omitted variables	excluded variables	More precise when referring to features deliberately removed during feature selection.
Yearly number of eligible posters	Annual number of eligible posters	More conventional for annual observations.
Military markers	Military-related indicators	Clearer terminology.
military publishers and organizational designers tag	military publisher-name markers and organizational designer-name markers	Corrects the grammatical structure and aligns with the feature names.
military themes and personnel tag	military-theme labels and military-personnel tags	Corrects parallelism and terminology.
Their peaks coincide with or precede heightened disputes	Their peaks coincide with or precede periods of heightened dispute activity	More natural expression.
indicating propaganda may signal military escalation via mass mobilizing	suggesting that propaganda may signal military escalation through mass mobilization	Fixes the awkward phrase *via mass mobilizing* and improves sentence structure.
Their relationship with military escalation, however, leaves room for more nuanced exploration	Their relationships with military escalation, however, remain unclear and warrant further investigation	More precise and conventional academic phrasing.

Mixed Evidence in Disputes	Mixed Evidence of Associations with Disputes	Clarifies the relationship being investigated.

3. LaTeX punctuation and formatting

There is one particular issue in your final figure caption.

Original:

```latex
\caption*{\small \textit{Note:}{Four annual visual summaries.
```

Corrected:

```latex
\caption*{\small \textit{Note:} Four annual visual summaries.
```

The extra pair of braces around the caption text is unnecessary. It is not necessarily a compilation error, but removing it makes the formatting consistent with your other figure notes.

Your use of `Figure~\ref{...}`, `Table~\ref{...}`, and `Appendix~\ref{...}` is appropriate. The nonbreaking spaces (`~`) correctly prevent line breaks between the reference labels and their numbers.

4. Additional academic writing observations

Three points are worth considering beyond grammar and punctuation.

1. "Declining Poster Production" versus archive coverage

Your figure apparently reports the number of eligible posters in your archive, rather than total historical poster production.

These are not necessarily equivalent. A decline in archived records could reflect changes in historical production, collection practices, digitization, or preservation.

A more cautious caption would be:

> Declining Availability of Propaganda Posters in the Archive

Similarly, you could revise the corresponding sentence to:

> Overall, however, a potential positive association can be observed: as China's involvement in militarized disputes declined, the number of military-related posters in the archive also appears to have decreased.

This avoids making an unsupported inference about actual historical production.

2. "Through mass mobilization"

Your current sentence implies a specific mechanism:

> Their peaks coincide with or precede periods of heightened dispute activity, suggesting that propaganda may signal military escalation through mass mobilization.

The grammar is correct, but the analysis presented here does not directly demonstrate a mass-mobilization mechanism.

A more cautious alternative would be:

> Their peaks coincide with or precede periods of heightened dispute activity, suggesting that military-related propaganda may contain signals relevant to forecasting military escalation.

This more directly reflects your research objective without presupposing a mechanism.

3. "Mixed Evidence" in the final figure title

Your original caption, *Distinct Visual Styles Across Historical Periods but Mixed Evidence in Disputes*, was somewhat ambiguous.

A more concise alternative to my revised version would be:

> Temporal Variation in Visual Features and Their Associations with Military Disputes

This better communicates the descriptive nature of your EDA and avoids implying that you have already established predictive relationships.

Overall assessment: The most substantial grammatical problem is the paragraph beginning with *Military markers show broadly similar patterns*, which contains inconsistent noun forms and the awkward expression *via mass mobilizing*. The remaining changes mainly concern tense consistency, academic phrasing, and minor LaTeX formatting. The structure and presentation of your EDA section are otherwise coherent.

你说：

Please check this text for grammatical and punctuation errors, reply in English: %
```
================================================
% 4. Problem Description
% ================================================
\section{Problem Description}

As discussed above, this project is essentially a regression task with a continuous outcome, which can be expressed as:

\begin{equation}
   \text{Military Escalation}_{t+1}
   = f(\text{Visual Features}_{t},
      \text{Poster Metadata}_{t},
      \text{Archive Labels}_{t})
\end{equation}

Our poster dataset provides the empirical foundation for this project. Previous studies of this collection have primarily focused on art history \citep{landsberger1994future, landsberger2013contextualising}, with limited applications in social science \citep{chen_xxxx}. To our knowledge, no study has systematically applied computational methods to this collection, particularly for forecasting.

For the outcome variable, we use the established MIDs dataset \citep{ward2007disputes, carroll2019prediction} rather than constructing an arbitrary alternative to better engage the scholarship.\footnote{This approach has been discussed with the TA.} Existing studies have demonstrated the predictive potential of neural networks \citep{beck2000improving} and unstructured diplomatic correspondence \citep{boussalis2022public}. Building on this scholarship, we seeks to explore the broader potential of multimodal data, particularly visual information, for improving military escalation forecasts.

Following the methods covered in the course, we will first estimate a linear regression model as an interpretable baseline. Drawing on the existing literature \citep{muchlinski2016comparing, hegre2021views2020}, we will further consider widely used machine learning models, including random forests and gradient boosting, to assess whether more flexible approaches improve overall performance.

Three additional technical considerations are particularly important:
```

\begin{itemize}
\item \textbf{Time-series structure:} Observations may violate the i.i.d.\ assumption. We therefore use chronological splits to prevent future information leakage.

\item \textbf{Choice of time window:} We initially predict one year ahead but acknowledge the arbitrariness. Alternative windows will be considered for robustness during model training.

\item \textbf{Missing values:} We will explore appropriate imputation strategies based on methods introduced later in the course.
\end{itemize}

% ==
% 5. Potential Impact (5 points)
% ==

\section{Potential Impact}

This project is an exploratory contribution to the long-standing challenge of war forecasting. Its primary audience is academic researchers in international relations and political communication. By examining whether propaganda posters improve predictive performance, we seek to demonstrate the potential of visual information as a non-traditional data source for conflict forecasting.

The project may also inform foreign policymaking. Although preliminary, the model could potentially help policymakers assess China's foreign policy through newly released posters. In particular, identifying which visual and textual features precede military escalation may provide additional insights to support broader policy assessments.
…
显示更多

ChatGPT 说：

Your **Problem Description** and **Potential Impact** sections are generally clear and grammatically sound. I found one definite subject–verb agreement error (*we seeks*), several awkward constructions, and a few places where the academic phrasing could be improved.

Below is a minimally revised version preserving your original meaning, structure, and LaTeX citations.

1. Corrected version

LaTeX

```
% ================================================
% 4. Problem Description
% ================================================
\section{Problem Description}

As discussed above, this project is essentially a regression task with a count outcome, which can be expres

\begin{equation}
    \text{Military Escalation}_{t+1}
    = f(\text{Visual Features}_{t},
        \text{Poster Metadata}_{t},
        \text{Archive Labels}_{t})
\end{equation}

Our poster dataset provides the empirical foundation for this project. Previous studies of this collection

For the outcome variable, we use the established MIDs dataset \citep{ward2007disputes, carroll2019predictio

Following the methods covered in the course, we will first estimate a linear regression model as an interpr
```

2. Specific corrections

Original	Suggested correction	Reason
a continuous outcome	a count outcome	Your dependent variable is the annual number of disputes, which is discrete rather than continuous.
which can be expressed as:	which can be expressed as follows:	More conventional introduction to a displayed equation.
in social science	in the social sciences	More conventional when referring to academic disciplines collectively.
rather than constructing an arbitrary alternative to better engage the scholarship	rather than constructing an arbitrary alternative, allowing us to engage more directly with the existing literature	Removes ambiguity about what *to better engage* modifies.
we seeks to explore	we seek to explore	Subject–verb agreement error.
acknowledge the arbitrariness	acknowledge the arbitrariness of this choice	Clarifies what is arbitrary.
a non-traditional data source	a nontraditional data source	Both are acceptable; *nontraditional* is common in American academic English.
the model could potentially help	the model could help	Removes redundancy between *could* and *potentially*.
assess China's foreign policy through newly released posters	assess China's foreign policy using newly released posters	More idiomatic phrasing for using posters as an information source.

3. Additional academic writing observations

1. Regression with a count outcome

The distinction between a continuous and count outcome is important.

Your target variable is the annual number of active militarized interstate disputes, meaning that:

$$Y_t \in \{0, 1, 2, \ldots\}$$

You can still treat this as a regression task and begin with linear regression. However, the dependent variable is technically a **count variable**, not a continuous variable.

Your revised opening sentence therefore correctly describes the task without requiring any change to your proposed models.

2. "An arbitrary alternative"

Although grammatically acceptable, *arbitrary* might unintentionally imply that constructing a different outcome measure would necessarily be methodologically unsound.

A more neutral alternative would be:

> For the outcome variable, we use the established MIDs dataset \citep rather than constructing a new measure, allowing us to engage more directly with the existing literature.

I slightly prefer this formulation for an academic paper.

3. "Multimodal data"

This term is appropriate if your project combines visual information with textual information or poster metadata.

However, because your selected predictors primarily consist of image-derived features, metadata, and archival annotations, you may want to distinguish *multimodal data* from *visual information* more explicitly in subsequent sections.

4. Potential policy applications

The following sentence is grammatically correct:

> Although preliminary, the model could help policymakers assess China's foreign policy using newly released posters.

There are two considerations:

First, *Although preliminary* logically modifies *the model*, so there is no dangling-modifier problem.

Second, your current dataset ends in 2013. Applying the model to newly released posters would therefore be a prospective extension rather than something directly established by your present analysis.

If you want to emphasize this distinction, an alternative is:

> Although still preliminary, this approach could eventually help policymakers assess China's foreign policy by analyzing newly released propaganda posters.

This preserves your original argument while making the prospective nature of the application clearer.

Overall assessment: The only definite grammatical error is *we seeks*. Most other revisions concern idiomatic academic English and clarity. The most important substantive correction is replacing *continuous outcome* with *count outcome*. Both sections otherwise read naturally and require no major restructuring.

你喜欢这种个性吗?